

Serological Testing Considerations in Syphilis

Non-treponemal agglutination tests, particularly in secondary syphilis, may yield weakly positive or negative results due to the prozone phenomenon, where excess antibodies interfere with the antigen-antibody reaction, preventing proper lattice formation or blocking the reaction.

Negative serological reactions can also occur in HIV-positive individuals or those with other forms of immunodeficiency. Additionally, false-negative results may be observed during the 'window' period, especially in early primary syphilis before seroconversion has occurred.

Serological Investigation of Cerebrospinal Fluid (CSF)

Serological testing for neurosyphilis is performed on whole cerebrospinal fluid.

- **VDRL-CSF** and **MPR-CSF** tests are highly specific but may lack sensitivity in certain forms of neurosyphilis.
- The **FTA-ABS-CSF** test is useful as an exclusion tool; a negative result strongly argues against neurosyphilis.
- Some laboratories employ treponemal **ELISA** assays for detecting specific antibodies in CSF. However, there is currently insufficient published evidence to validate the diagnostic accuracy of these assays for neurosyphilis. Results from such tests should be interpreted with caution and not used alone to confirm or exclude neurosyphilis.

Indications for Serological Testing

General Screening Recommendations

Syphilis screening should be conducted using one of the following tests: **MPR**, **RPR**, **ELISA**, **TPHA**, or **TPPA**, depending on local epidemiology and laboratory capabilities.

- In populations with **low prevalence** (e.g., general hospital inpatients, outpatient clinics, routine health examinations), **treponemal screening** (e.g., ELISA, TPHA, FTA) is preferred. This allows identification of individuals previously or currently infected. Positive results should be confirmed with a **non-treponemal test** (RPR/MPR) to assess disease activity.
- In populations with **high prevalence** (e.g., sex workers, prisoners), initial screening should use a **non-treponemal test** (RPR/MPR). All reactive results must be confirmed with a **treponemal test**.

Special Clinical Settings

- In patients presenting to **ophthalmology**, **psychoneurology**, or **cardiology** departments, where late syphilis may be suspected, testing should combine a **non-treponemal test** with a treponemal test such as **ELISA**, **TPHA**, **FTA**, or **Western blot**, based on laboratory availability.
- For **pregnant women** and **blood donors**, screening must include both a **non-treponemal test** and a treponemal test (**ELISA**, **TPHA**, **FTA**, **Western blot**, or **TPI**), depending on available resources.
- In high-prevalence countries, pregnant women should undergo serological screening **three times**: at first antenatal visit, at **18–20 weeks**, and at **32 weeks** of gestation.

Clinical Presentations

- **Suspected primary syphilis:** Test with a **non-treponemal test** (RPR/MPR). All positive results should be confirmed with a treponemal test (**FTA, TPHA, modified Western blot**), or **ELISA** if validated and available.
- **Suspected secondary syphilis:** Perform a **non-treponemal test** in combination with a treponemal test (**ELISA, TPHA, or FTA**).
- **Suspected late (tertiary) syphilis** (including cardiovascular, musculoskeletal, or visceral involvement): Use a **non-treponemal test** plus **two treponemal tests**, such as:
 - **ELISA or FTA + TPHA**
 - **ELISA or FTA + TPI**
- **Asymptomatic individuals with recent sexual or close household exposure** to contagious syphilis: Evaluate as suspected **primary syphilis**.
- **Suspected latent syphilis** (no clinical signs): Test with a **non-treponemal test** and **two treponemal tests**. Acceptable combinations include:
 - **ELISA or FTA + TPHA**
 - **ELISA or FTA + TPI**